



SISTEM PERSAMAAN

Tentukan solusi (x, y) dari sistem persamaan dibawah ini!

I. Tentukan solusi dari SPLDV berikut dengan metode substitusi!

$$\begin{array}{l} x - 2(3) = -4 \\ x - 6 = -4 \\ x = -4 + 6 \\ x = 2 \end{array}$$

a. $\begin{cases} 2x + 3y = 13 \\ x - 2y = -4 \end{cases}$ $\begin{array}{l} 2x + 3y = 13 \\ 2x - 4y = -8 \\ \hline 7y = 21 \\ y = 3 \end{array}$ $\begin{array}{l} x - 2(3) = -4 \\ x - 6 = -4 \\ x = 2 \end{array}$

b. $\begin{cases} 2x + 3y = 13 \\ 4x - y = 5 \end{cases}$ $\begin{array}{l} 2x + 3y = 13 \\ 4x - y = 5 \\ \hline 4x - 3x = 5 - 6 \\ x = -1 \end{array}$ $\begin{array}{l} 2(-1) + 3y = 13 \\ -2 + 3y = 13 \\ 3y = 15 \\ y = 5 \end{array}$

c. $\begin{cases} x + 4y = 13 \\ 2x - y = 53 \end{cases}$ $\begin{array}{l} x + 4y = 13 \\ 2x - y = 53 \\ \hline 2x + 8y = 26 \\ 2x - y = 53 \\ \hline 9y = -27 \\ y = -3 \end{array}$ $\begin{array}{l} x + 4(-3) = 13 \\ x - 12 = 13 \\ x = 25 \end{array}$

d. $\begin{cases} 5x - 3y = 16 \\ 3x - 4y = 14 \end{cases}$ $\begin{array}{l} 5x - 3y = 16 \\ 3x - 4y = 14 \\ \hline 20x - 12y = 64 \\ 15x - 12y = 42 \\ \hline 5x = 22 \\ x = 4.4 \end{array}$

e. $\begin{cases} 4x + 2y = 5 \\ 2x + 3y = 7.5 \end{cases}$ $\begin{array}{l} 4x + 2y = 5 \\ 2x + 3y = 7.5 \\ \hline 4x + 2y = 5 \\ 4x + 6y = 15 \\ \hline -4y = -10 \\ y = 2.5 \end{array}$ $\begin{array}{l} 4x + 2(2.5) = 5 \\ 4x + 5 = 5 \\ 4x = 0 \\ x = 0 \end{array}$

f. $\begin{cases} 2x + 3y = 7 \\ 2x - 4y = 14 \end{cases}$ $\begin{array}{l} 2x + 3y = 7 \\ 2x - 4y = 14 \\ \hline 7y = -7 \\ y = -1 \end{array}$ $\begin{array}{l} 2x + 3(-1) = 7 \\ 2x - 3 = 7 \\ 2x = 10 \\ x = 5 \end{array}$

g. $\begin{cases} 4x + y = -14 \\ x - 2y = 1 \end{cases}$ $\begin{array}{l} 4x + y = -14 \\ x - 2y = 1 \\ \hline 4x + y = -14 \\ 4x - 8y = 4 \\ \hline 9y = -18 \\ y = -2 \end{array}$ $\begin{array}{l} 4x + (-2) = -14 \\ 4x - 2 = -14 \\ 4x = -12 \\ x = -3 \end{array}$

II. Tentukan solusi dari SPLDV berikut dengan metode eliminasi!

a. $\begin{cases} 4x - 5y = 7 \\ 2x + 3y = 4 \end{cases}$ $\begin{array}{l} 4x - 5y = 7 \\ 2x + 3y = 4 \\ \hline 2x - 11y = 3 \\ 2x + 3y = 4 \\ \hline -14y = -1 \\ y = \frac{1}{14} \end{array}$ $\begin{array}{l} 4x - 5(\frac{1}{14}) = 7 \\ 4x - \frac{5}{14} = 7 \\ 4x = 7 + \frac{5}{14} \\ 4x = \frac{103}{14} \\ x = \frac{103}{56} \end{array}$

b. $\begin{cases} 3x - y = 4 \\ 2x + 2y = 8 \end{cases}$ $\begin{array}{l} 3x - y = 4 \\ 2x + 2y = 8 \\ \hline 6x - 2y = 8 \\ 2x + 2y = 8 \\ \hline 4x = 16 \\ x = 4 \end{array}$ $\begin{array}{l} 3(4) - y = 4 \\ 12 - y = 4 \\ -y = -8 \\ y = 8 \end{array}$

c. $\begin{cases} 2x + 3y = 13 \\ 4x - y = 5 \end{cases}$ $\begin{array}{l} 2x + 3y = 13 \\ 4x - y = 5 \\ \hline 2x + 3y = 13 \\ 4x - y = 5 \\ \hline -2x + 4y = 8 \\ -2x + 4y = 8 \\ \hline 0 = 0 \end{array}$ $\begin{array}{l} 2x + 3y = 13 \\ 4x - y = 5 \\ \hline 2x + 3y = 13 \\ 4x - y = 5 \\ \hline -2x + 4y = 8 \\ -2x + 4y = 8 \\ \hline 0 = 0 \end{array}$

d. $\begin{cases} 3x + 2y = 12 \\ 4x - 3y = 1 \end{cases}$ $\begin{array}{l} 3x + 2y = 12 \\ 4x - 3y = 1 \\ \hline 12x + 6y = 36 \\ 16x - 9y = 4 \\ \hline -4x + 15y = 32 \\ -4x + 15y = 32 \\ \hline 0 = 0 \end{array}$ $\begin{array}{l} 3x + 2y = 12 \\ 4x - 3y = 1 \\ \hline 12x + 6y = 36 \\ 16x - 9y = 4 \\ \hline -4x + 15y = 32 \\ -4x + 15y = 32 \\ \hline 0 = 0 \end{array}$

e. $\begin{cases} 4x + 6y = 34 \\ 2x + 6y = 26 \end{cases}$ $\begin{array}{l} 4x + 6y = 34 \\ 2x + 6y = 26 \\ \hline 2x = 8 \\ x = 4 \end{array}$ $\begin{array}{l} 4(4) + 6y = 34 \\ 16 + 6y = 34 \\ 6y = 18 \\ y = 3 \end{array}$

III. Tentukan solusi dari SPLDV berikut dengan metode campuran!

a. $\begin{cases} 3x + 2y = 12 \\ 4x + 3y = 17 \end{cases}$ $\begin{array}{l} 3x + 2y = 12 \\ 4x + 3y = 17 \\ \hline 12x + 8y = 48 \\ 16x + 12y = 68 \\ \hline -4x - 4y = -20 \\ -4x - 4y = -20 \\ \hline 0 = 0 \end{array}$ $\begin{array}{l} 3x + 2y = 12 \\ 4x + 3y = 17 \\ \hline 12x + 8y = 48 \\ 16x + 12y = 68 \\ \hline -4x - 4y = -20 \\ -4x - 4y = -20 \\ \hline 0 = 0 \end{array}$

b. $\begin{cases} 4x - 2y = 2 \\ 5x + 4y = 35 \end{cases}$ $\begin{array}{l} 4x - 2y = 2 \\ 5x + 4y = 35 \\ \hline 4x - 2y = 2 \\ 5x + 4y = 35 \\ \hline -x + 6y = -33 \\ -x + 6y = -33 \\ \hline 0 = 0 \end{array}$ $\begin{array}{l} 4x - 2y = 2 \\ 5x + 4y = 35 \\ \hline 4x - 2y = 2 \\ 5x + 4y = 35 \\ \hline -x + 6y = -33 \\ -x + 6y = -33 \\ \hline 0 = 0 \end{array}$

